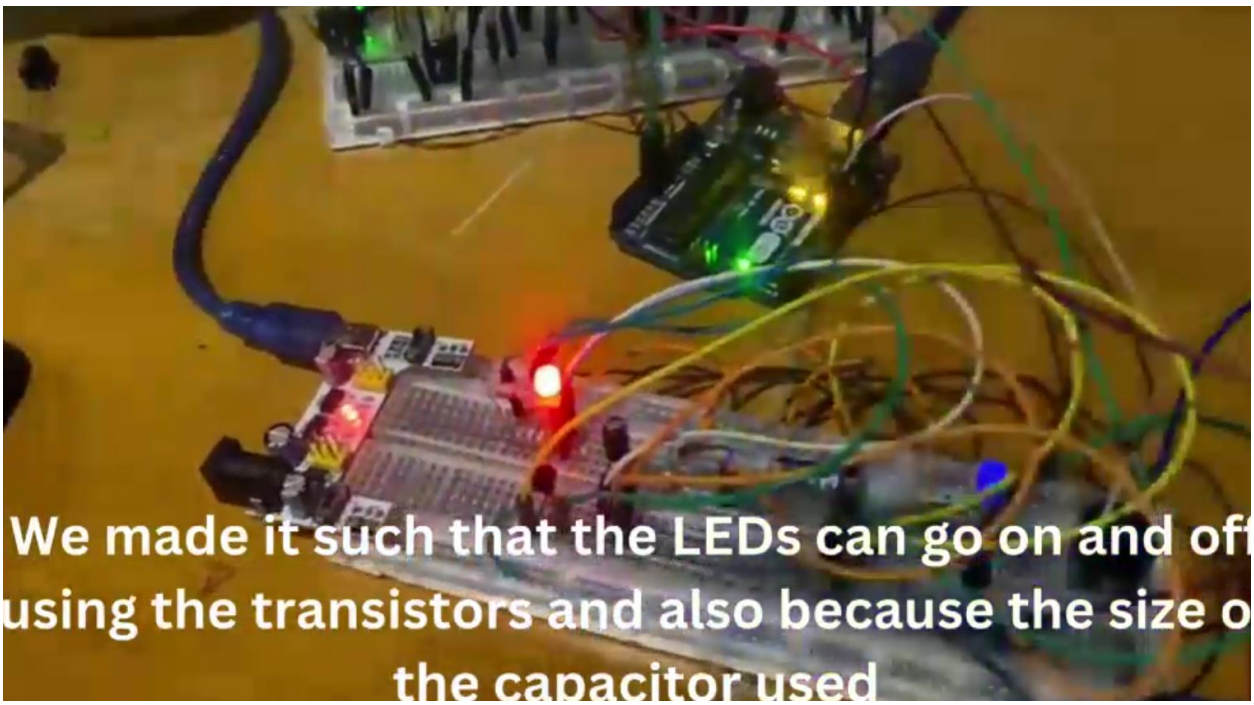
Sean and Shadrack

Circuits and electronics

AN INTRUDER ALARM SYSTEM USING AN ASABLE MULTIVIRATOR



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Abstract: This project involves the development of a simple yet effective object detection and alert system using an astable multivibrator, an infrared (IR) sensor, an Arduino Uno, and a buzzer. The astable multivibrator circuit constructed using two 2N3904 transistors, capacitors, resistors, and LEDs, functions as an oscillator, producing a continuous square wave output that alternately blinks the LEDs. The design is simple, cost-effective, and suitable for applications such as entry detection, proximity alerts, or any scenario requiring immediate feedback upon object detection.

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Introduction:

This project focuses on designing and implementing an object detection and alert system, utilizing an astable multivibrator circuit, an infrared (IR) sensor, and an Arduino Uno microcontroller. The primary objective is to develop a system that can detect the presence of an object and provide both visual and auditory alerts, showcasing the practical application of fundamental electronic components and microcontroller programming. The system's design emphasizes simplicity, cost-effectiveness, and ease of implementation, making it an ideal project for educational purposes, hobbyists, and entry-level applications. The combination of analog electronic circuits with digital control provided by the Arduino Uno allows for a flexible and expandable platform, with potential future enhancements such as integrating additional sensors, wireless communication, or more sophisticated alert mechanisms.

In the following sections, the paper will delve into the detailed design, component selection, circuit implementation, and programming logic that underpin the successful operation of this object detection and alert system. Through this project, the foundational concepts of electronics and microcontroller-based systems are explored, providing valuable insights into the creation of practical, real-world applications.

Methodology:

IR Sensor: The IR module is used to detect the presence of an object or a certain IR signal. It has a 5V power supply, ground, and an output pin connected to the Arduino (A0). This module sends a signal to the Arduino when it detects an object or signal.

Buzzer: The buzzer is used to provide an audible alert when the IR module detects something. It is connected to one of the digital output pins on the Arduino.

Capacitors C1 and C2: These components are connected to the bases of the transistors and are responsible for the charging and discharging processes that cause the circuit to switch states.

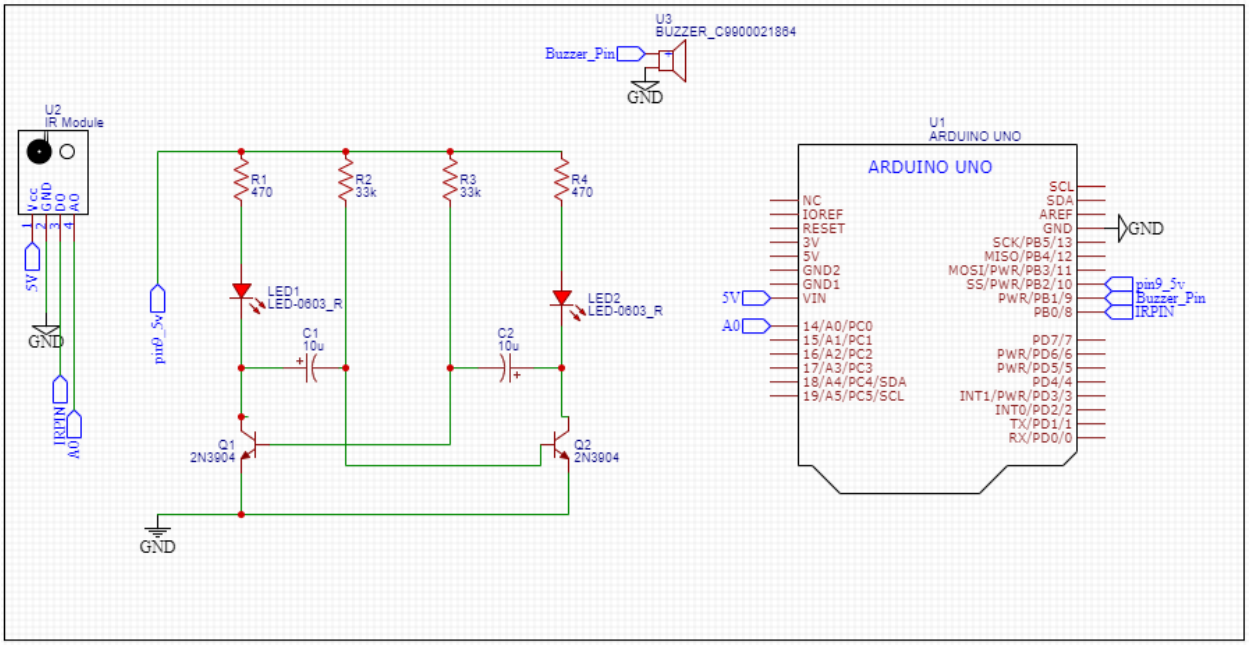
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Resistors: These limit the current to the transistors and set the charging time of the capacitors, which controls the frequency of the output

Transistors: The circuit uses two *NPN transistors* (like the ones in the image) that are connected in a way that one transistor turning on causes the other to turn off.

Arduino UNO: The Arduino Uno microcontroller receives the signal from the IR module through the analog pin (A0). The Arduino controls the buzzer (U3) through the Buzzer_Pin, likely using a digital output pin to turn the buzzer on and off depending on the input from the IR module.

LEDs: These LEDs indicate the oscillation of the circuit, flashing alternately as the circuit oscillates.



Results and Observations:

Our circuits operate as an astable multivibrator. The LED on the left side is lit when the transistor on the left side (Q1) is ON. The LED on the right side is lit when the transistor on the right side (Q2) is ON. Resistors R1 and R4 are only there to set the current through

the LEDs. Which means the remaining six components make up the oscillator: Q1, Q2, C1, C2, R2, and R3.

The transistor acts as a switch. It needs 0.7V on the base to turn ON. When it is ON, its top collector connects down to its emitter so that current can flow through it. This also means that the top pin has the same voltage as the bottom pin when the transistor is on. When the transistor is OFF, there are no connections between the top pin and the bottom pin, so no current can flow.

When Q1 is on, it allows current to flow from the collector to the emitter, which turns on LED1 (L1). Capacitor C2 begins to charge through resistor R3. As C2 charges, the voltage across it gradually increases. This voltage is applied to the base of Q2. When the voltage across C2 reaches a certain threshold, it forward-biases Q2, turning it on. As Q2 turns on, Q1 begins to turn off because the charge on C1 (which was previously charging) starts discharging, reducing the base voltage of Q1. With Q2 now on and Q1 off, the current flows through R4 and turns on LED2 (L2). Now, C1 starts charging through R2. This charge buildup across C1 will eventually forward-bias Q1, turning it back on and repeating the cycle.

We connected our astable multivibrators power supply to an Arduino pin so we programmed it for power to be supplied only when there’s an object.
We observed that when the IR sensor detects an objects, the LEDs come on and the buzzer keeps sounding till the object is no more.

Limitations and conclusion:

Even though we successfully created the intruder alarm system using an astable multivibrator and IR sensor, we noticed a few limitations. First, the IR sensor's range isn't great, so it struggles to detect objects that are far away or when there's a lot of light in the environment. Also, since we're using a buzzer for alerts, it might not be the best in noisy

places or situations where a quieter alarm is needed. Our design also doesn’t support remote monitoring or notifications, which would make it more useful for real-world security. Finally, because we used basic components and didn’t include advanced processing, the system can’t really tell the difference between different objects, which might lead to some false alarms.

In conclusion, our intruder alarm system shows how basic electronics and microcontroller programming can come together through an astable multivibrator, IR sensor, and Arduino Uno. We met our main goal of creating a simple and affordable solution for detecting objects and providing alerts. However, the limitations we encountered indicate that there's room for improvement. We could work on extending the system's range, making it less prone to false alarms, and adding more advanced alert features. In the future, we could also think about incorporating better sensors, enabling remote monitoring, and tailoring the design for specific needs. Overall, this project has been a great learning experience, demonstrating how basic electronic components and microcontrollers can be used in practical, real-world applications.